Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Tosini, Gianluca Ph.D.		
Project Title:	Preventing Desynchronization of the Circadian System in Long-Term Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30310-1495	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2004	**End Date:**	08/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fukuhara, Chiaki (Morehouse School of Medicine)		
Grant/Contract No.:	NCC 9-58-HPF00407		
Performance Goal No.:			
Performance Goal Text:			

Many biochemical, physiological and behavioral parameters exhibited by organisms show daily fluctuations and most of these daily rhythms persist in constant conditions, thus, demonstrating that they are driven by endogenous oscillators. The rhythms that persist in constant conditions with a period close to 24 hours are called circadian rhythms. One the most important aspects of space flight is the absence of geophysical 24-h cycles, which, of course, affects the overall temporal organization of the organisms. In the case of long-duration manned space flight, it is crucial to understand how the whole circadian system would react and behave in such circumstances.
We discovered that exposing rats to constant dim light for 60 days may induce spontaneous internal desynchronization in a few animals. In the present research proposal we further examined this phenomenon by investigating the physiological consequences of spontaneous internal desynchronization and to develop countermeasures to obviate the occurrence of internal desynchronization in animals exposed to constant dim light. Spontaneous internal

Task Description:	desynchronization has profound effects on the capability of the organisms to perform (mentally and physically) and to remain healthy. Recent studies have shown that a subset of retinal ganglion cells (RGCs) innervating the SCN are directly photosensitive and able to convert electromagnetic radiation into neural signals. Melanopsin, a photopigment based on vitamin A, was found in these RGCs and is the strongest candidate for the circadian photopigment within these cells. The spectral sensitivity of these RGC peak around 474 nm. It is now believed that these RGCs provide the main light input to the circadian clock. In the last year of funding we have developed a series of light blue narrow-band light-emitting diodes (LEDs)with an emission peak at 476 nm that should be very effective in stimulating the photosensitive retinal ganglion cells. Indeed, our preliminary data indicate that these LEDs may be a valid countermeasure to prevent disfunctions of the circadian system that may occur in the space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project has provided important data on the physiological mechanisms that are involved in the maintenance of the synchronization among the different circadian oscillators. The data that will be generated by this project will allow the design of specific treatments (pharmacological or environmental) that will prevent the desynchronization of the circadian system. Such treatments will then be employed in the treatment of circadian dysfunctions due to a pathological (or environmental) condition that may affect some individuals. Furthermore, the LEDs that we have develop may be useful in the treatment of several pathologies such as seasonal affective disorders and sleep disorders.
Task Progress:	During the last year we have tested the efficacy of our new blue narrow-band light-emitting diodes (LEDs). Our data indicate that these LEDs are very effective in modulating some circadian parameters (melatonin suppression and phase-shift of the locomotor activity). Preliminary data support the safety of these LEDs.
Bibliography Type:	Description: (Last Updated: 02/21/2008)
Articles in Peer-reviewed Journals	Tosini G, Davidson AJ, Fukuhara C, Kasamatsu M, Castanon-Cervantes O. "Localization of a circadian clock in mammalian photoreceptors." FASEB J. 2007 Dec;21(14):3866-71. Epub 2007 Jul 9. PMID: 17621597 , Dec-2007
Articles in Peer-reviewed Journals	Tosini G, Kasamatsu M, Sakamoto K. "Clock gene expression in the rat retina: effects of lighting conditions and photoreceptor degeneration." Brain Res. 2007 Jul 23;1159:134-40. PMID: 17560558 , Jul-2007
Articles in Peer-reviewed Journals	Tosini G, Aguzzi J, Bullock NM, Liu C, Kasamatsu M. "Effect of photoreceptor degeneration on circadian photoreception and free-running period in the Royal College of Surgeons rat." Brain Res. 2007 May 7;1148:76-82. PMID: 17382912 , May-2007
Awards	Tosini G. "Selected as 2008 Copernicus Visiting Scientist by University of Ferrara, August 2007." Aug-2007
Papers from Meeting Proceedings	Nicol D, Li N, Park E- U, Tosini G, Ferguson IT. "Imitating broadband diurnal light variations using solid state light light sources." 1st International Conference on Display LEDs (ICDL 2007), Seoul, Korea, Jan 31 - Feb 2, 2007. Proceedings of the 1st International Conference on Display LEDs (ICDL 2007), Seoul, Korea, Jan 31 - Feb 2, 2007. , Feb-2007